

Lecture 14: Maps

Heather Mattie, PhD

October 16, 2025



Recipe of the Day!

Butternut Squash Pasta with Walnuts and Sage



~ when you realize tomorrow
is Friday~

Announcements

- Homework #3 due Friday, October 24 by 11:59pm
- We will finish lab 4 during lab this week, any extra time will be used as office hours
- Mid-course evaluation is still live!

Coding question of the day

Using the **murders** dataset, create a plot of murder rates with 4 boxplots next to each other - one boxplot for each region.

Order the boxplots by median rate (you will need the **reorder** function for this).

Make sure the boxplots are different colors, the axes are informative, and you add a title.

Remove the legend by adding this layer to your plot:
guides(fill = "none")

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data("murders")
```

Comparing distributions

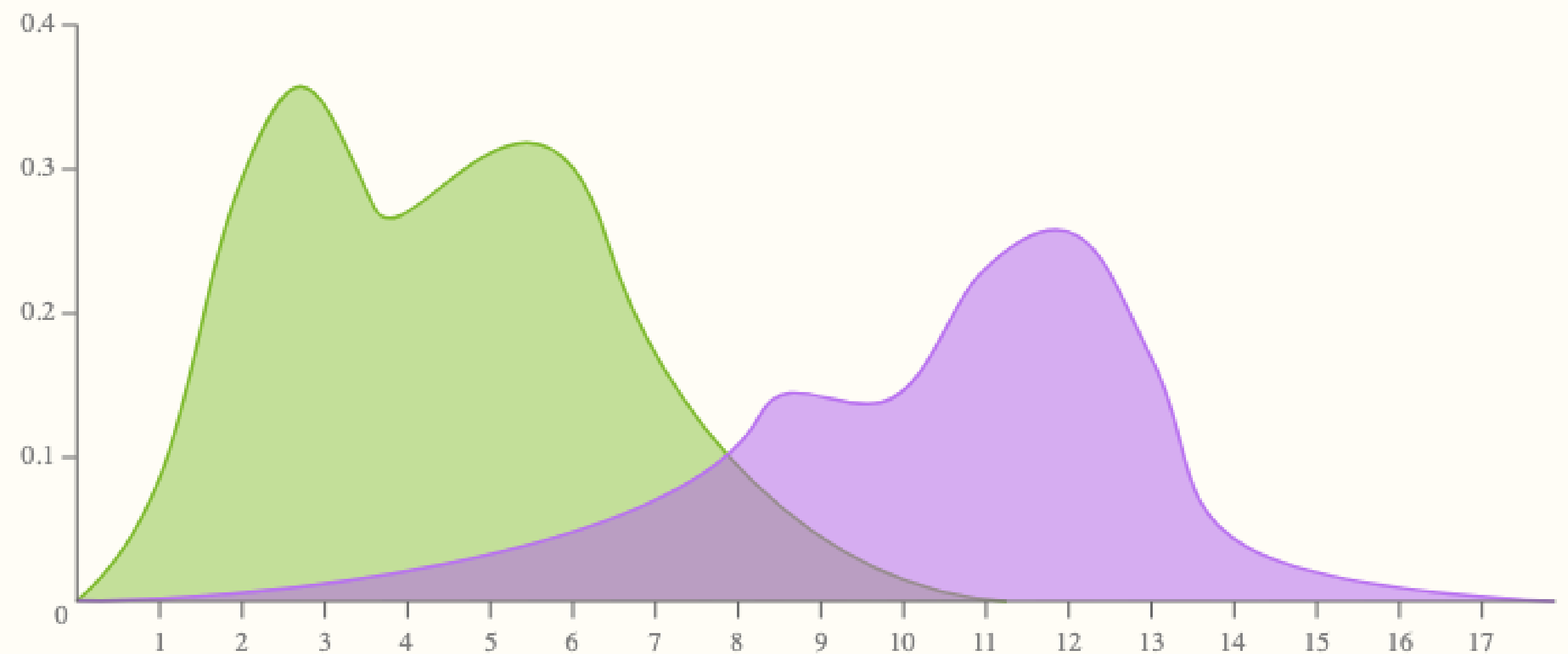
Smooth density plots

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Smooth density plots

Smooth density plots estimate the probability distribution of a continuous variable

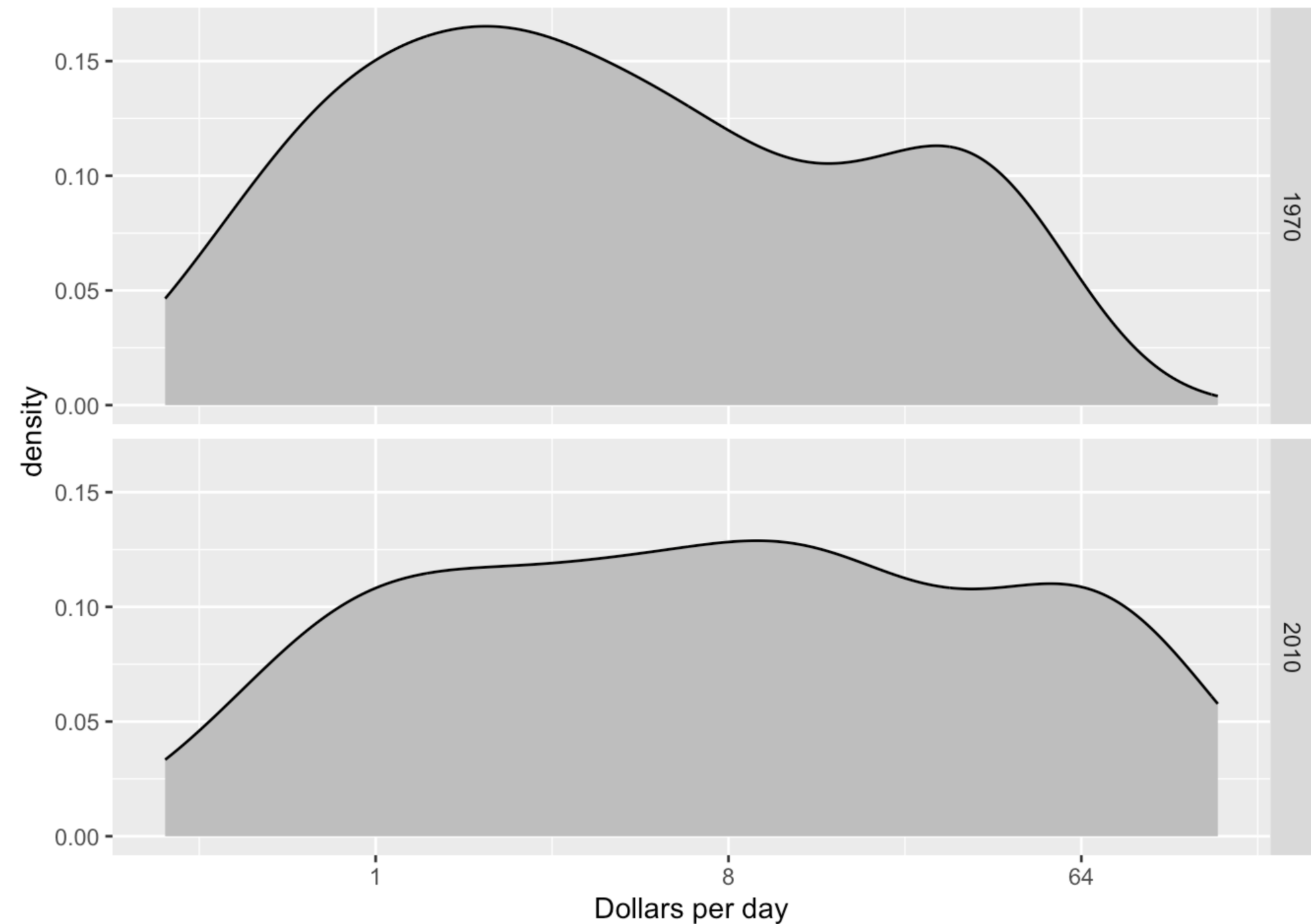
- A smoothed version of a histogram
- Useful for visualizing distribution shape
- Useful for comparing groups



Income distributions

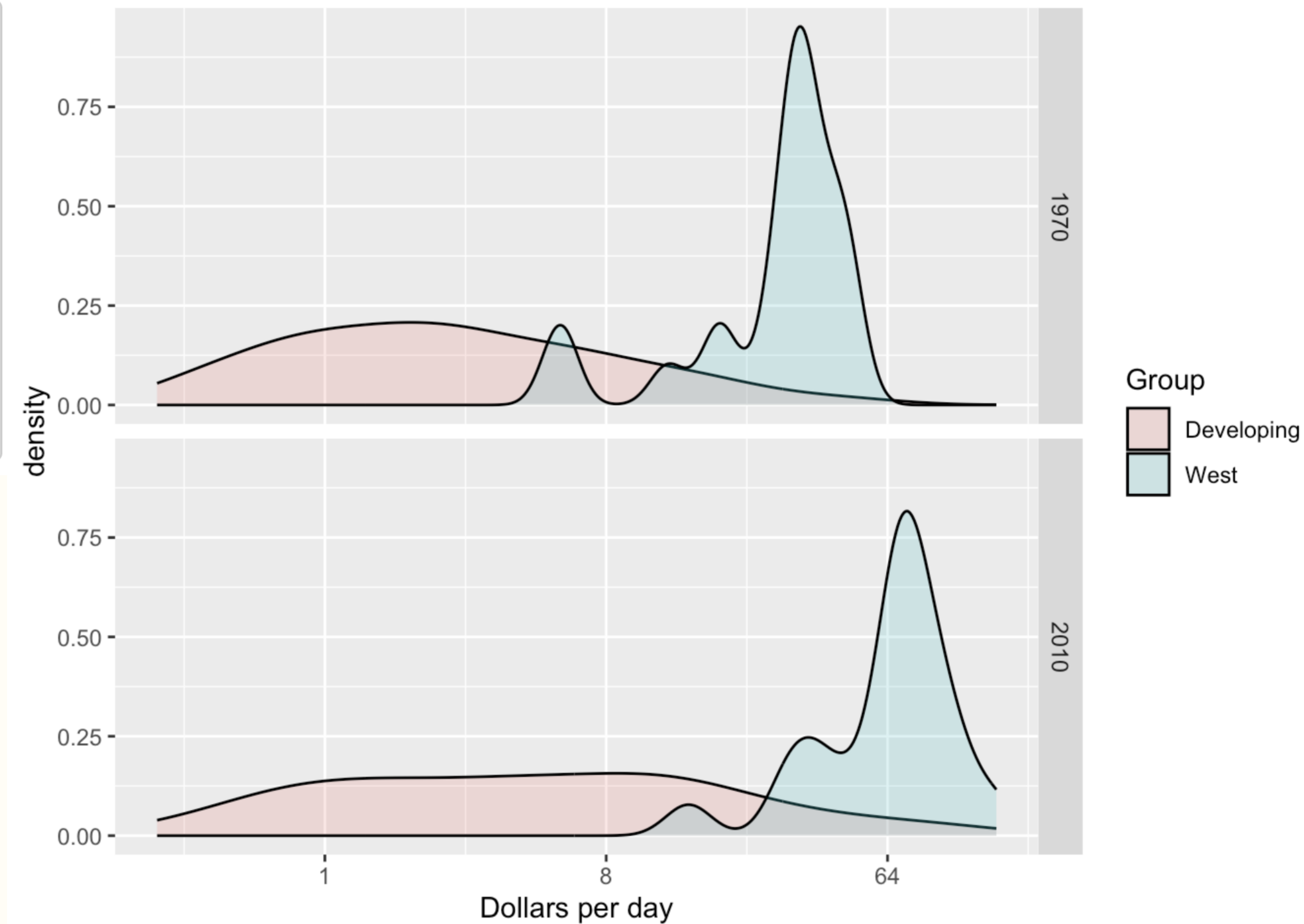
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
          country %in% country_list) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_density(fill = "grey") +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  facet_grid(year~.)
```

With this plot we can see that some countries have shifted to the right, closing the income gap – but we don't know **which countries** these are



Groups

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2) +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

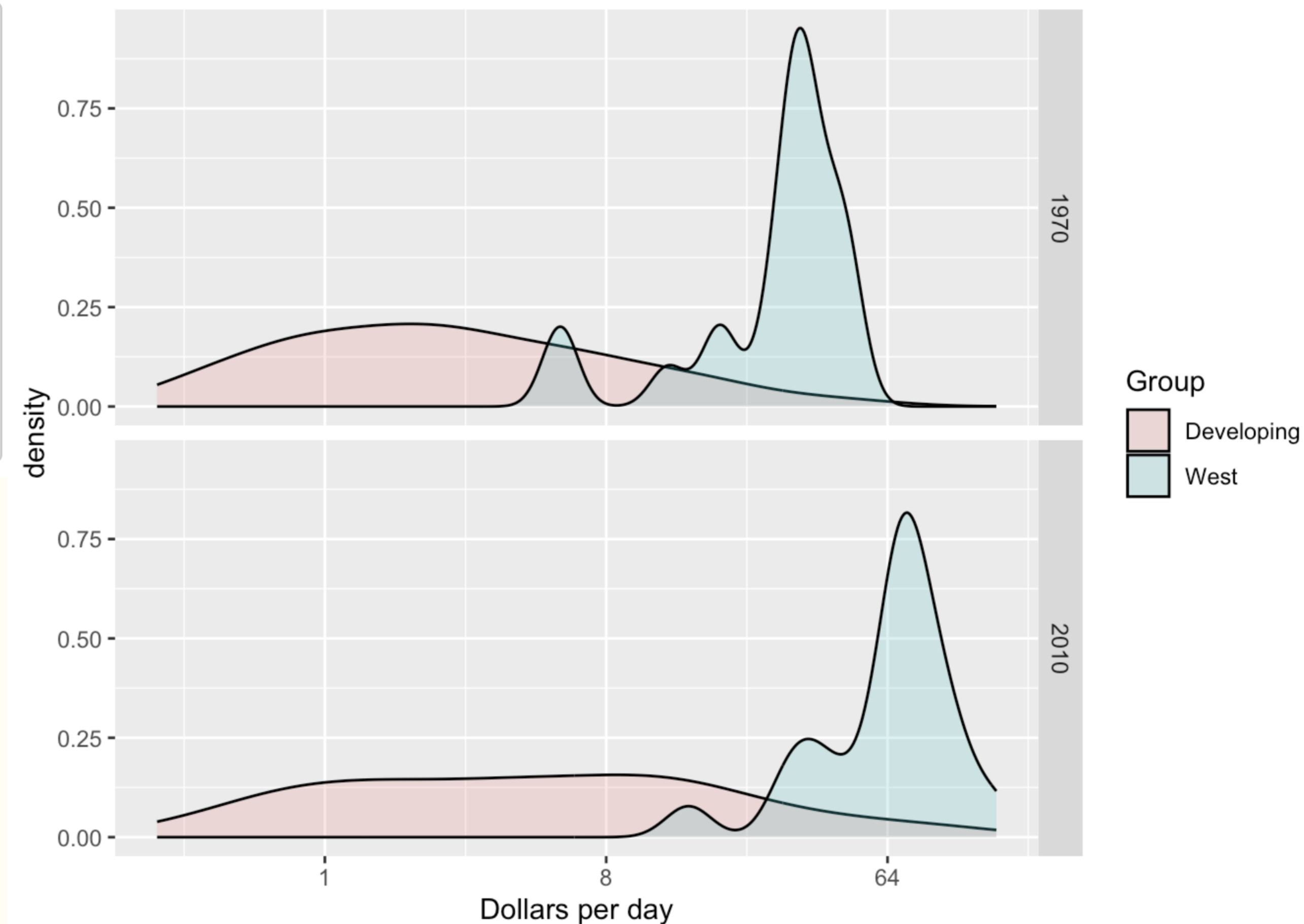


Groups

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2) +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

The default is for the area under a smooth density plot to be **equal to 1**

This is not ideal in this context since there are 21 countries in our West group and 87 in our Developing group

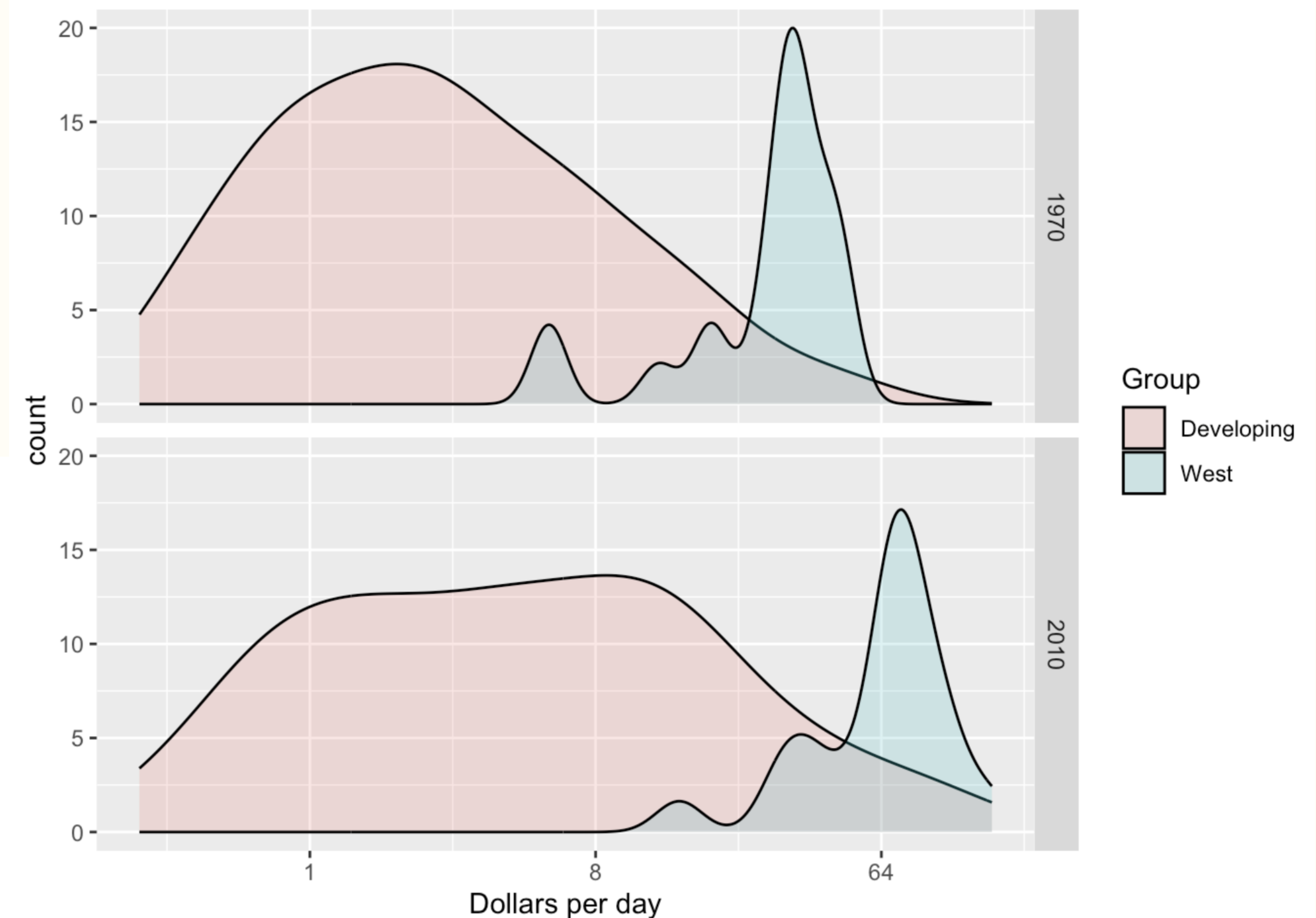


Accessing computed variables

Multiplying the y-axis values by the size of a group makes the area proportional to the number of countries in that group

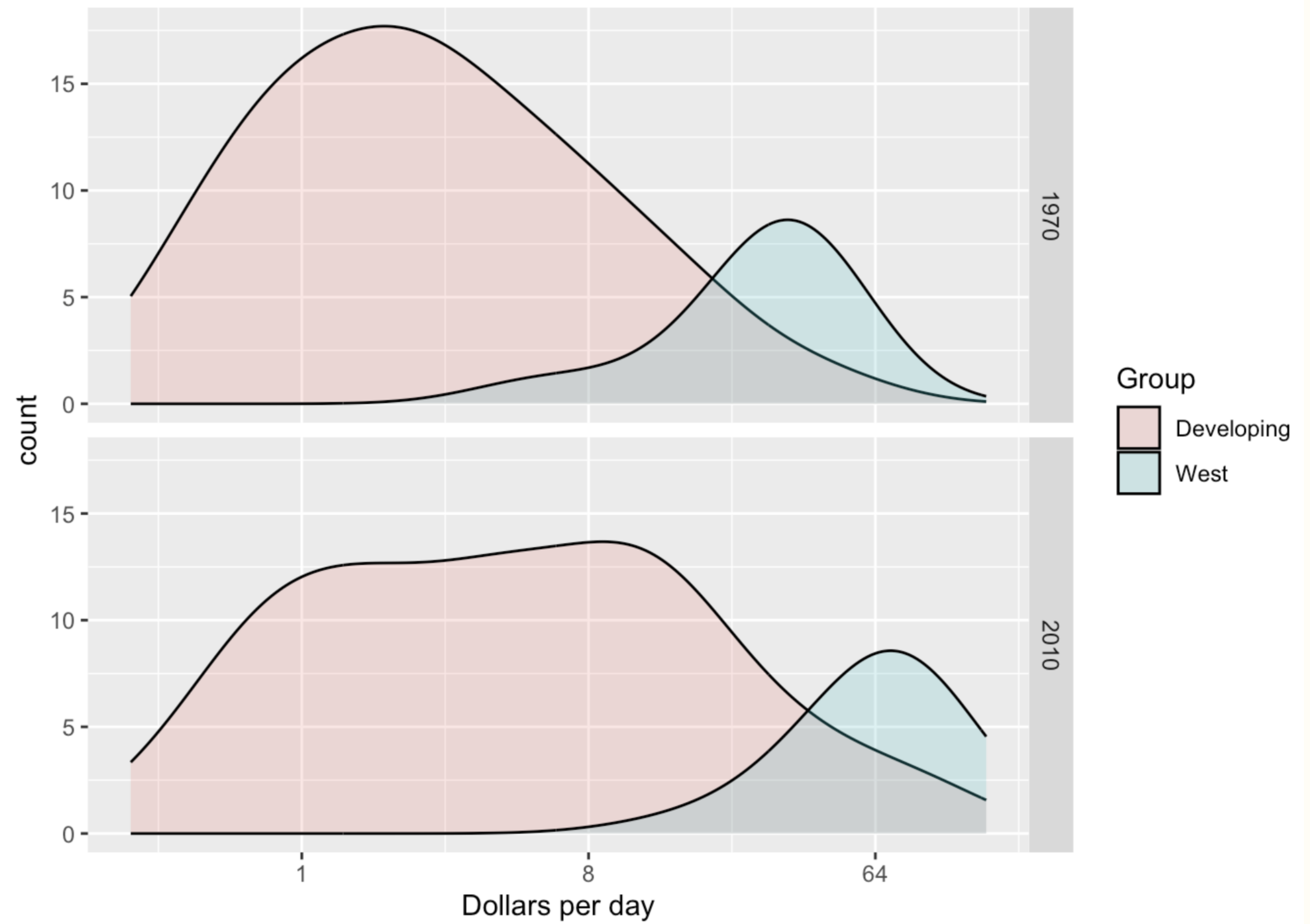
- **y = after_stat(count)**

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count), fill = group)) +  
  geom_density(alpha = 0.2) +  
  facet_grid(year ~ .) +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



Smoother densities (**bw**)

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count), fill = group)) +  
  geom_density(alpha = 0.2, bw = 0.75) +  
  facet_grid(year ~ .) +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



case_when

We can make this plot more informative by having more group categories

- The **case_when** function can create a new group variable efficiently

```
gapminder <- gapminder %>%  
  mutate(group = case_when(  
    region %in% west ~ "West",  
    region %in% c("Eastern Asia", "South-Eastern Asia") ~ "East Asia",  
    region %in% c("Caribbean", "Central America", "South America") ~ "Latin America",  
    continent == "Africa" & region != "Northern Africa" ~ "Sub-Saharan Africa",  
    TRUE ~ "Others"))
```

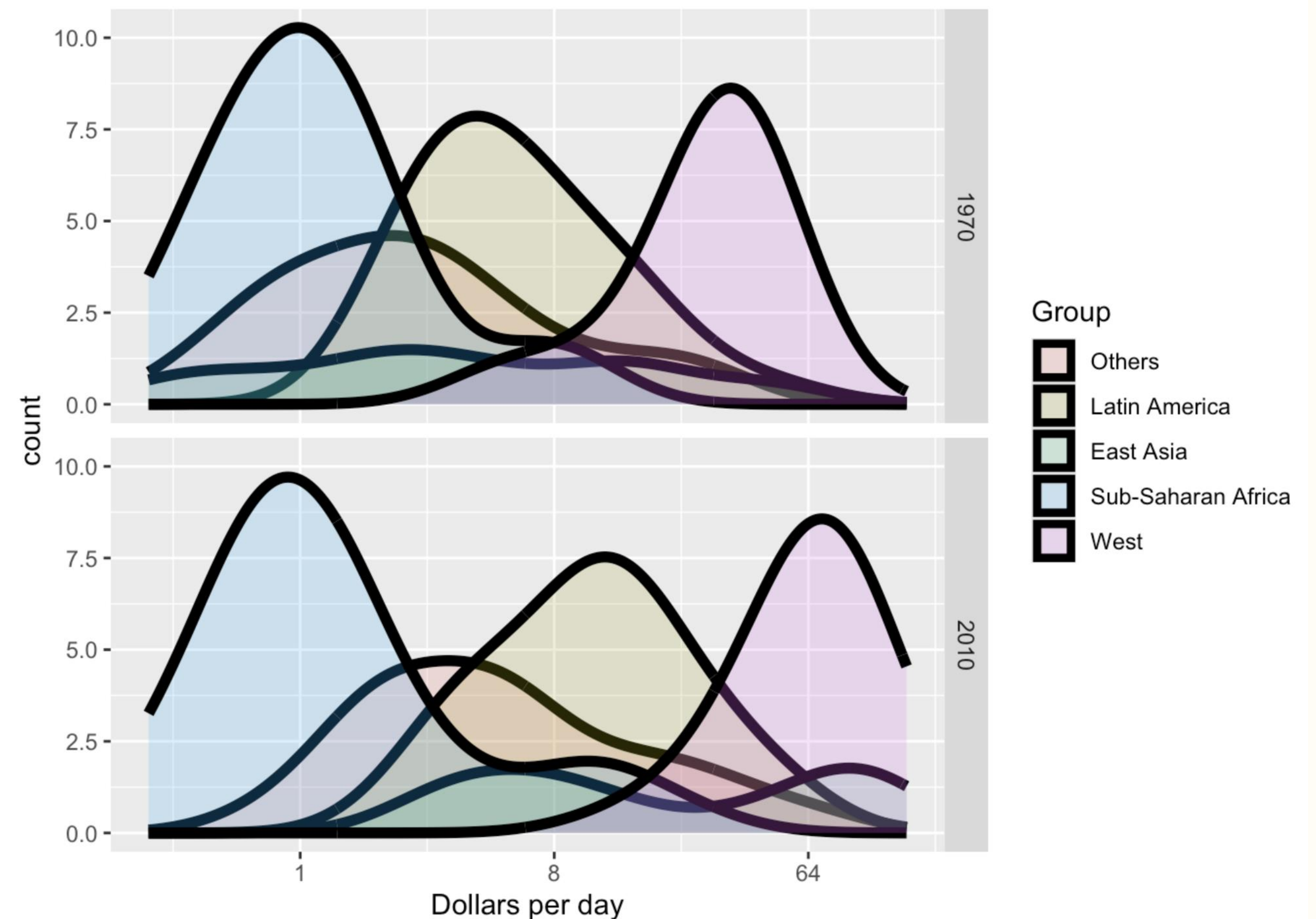
```
gapminder %>% select(region, group) %>% head()
```

##	region	group
## 1	Southern Europe	West
## 2	Northern Africa	Others
## 3	Middle Africa	Sub-Saharan Africa
## 4	Caribbean	Latin America
## 5	South America	Latin America
## 6	Western Asia	Others

linewidth

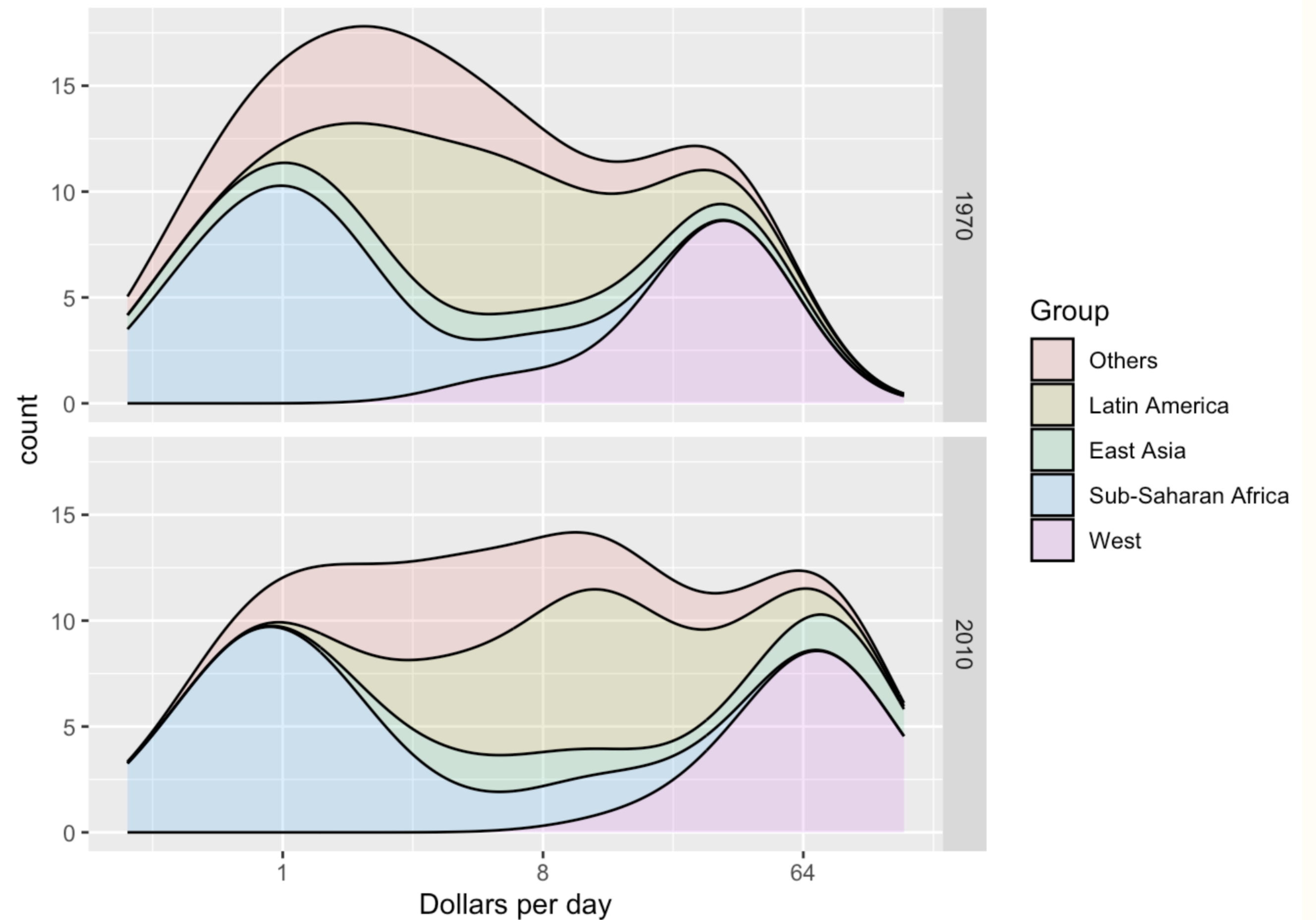
```
gapminder <- gapminder %>%  
  mutate(group = factor(group,  
    levels = c("Others", "Latin America",  
              "East Asia", "Sub-Saharan Africa",  
              "West")))
```

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
    country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
    fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, linewidth = 2) +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



position = "stack"

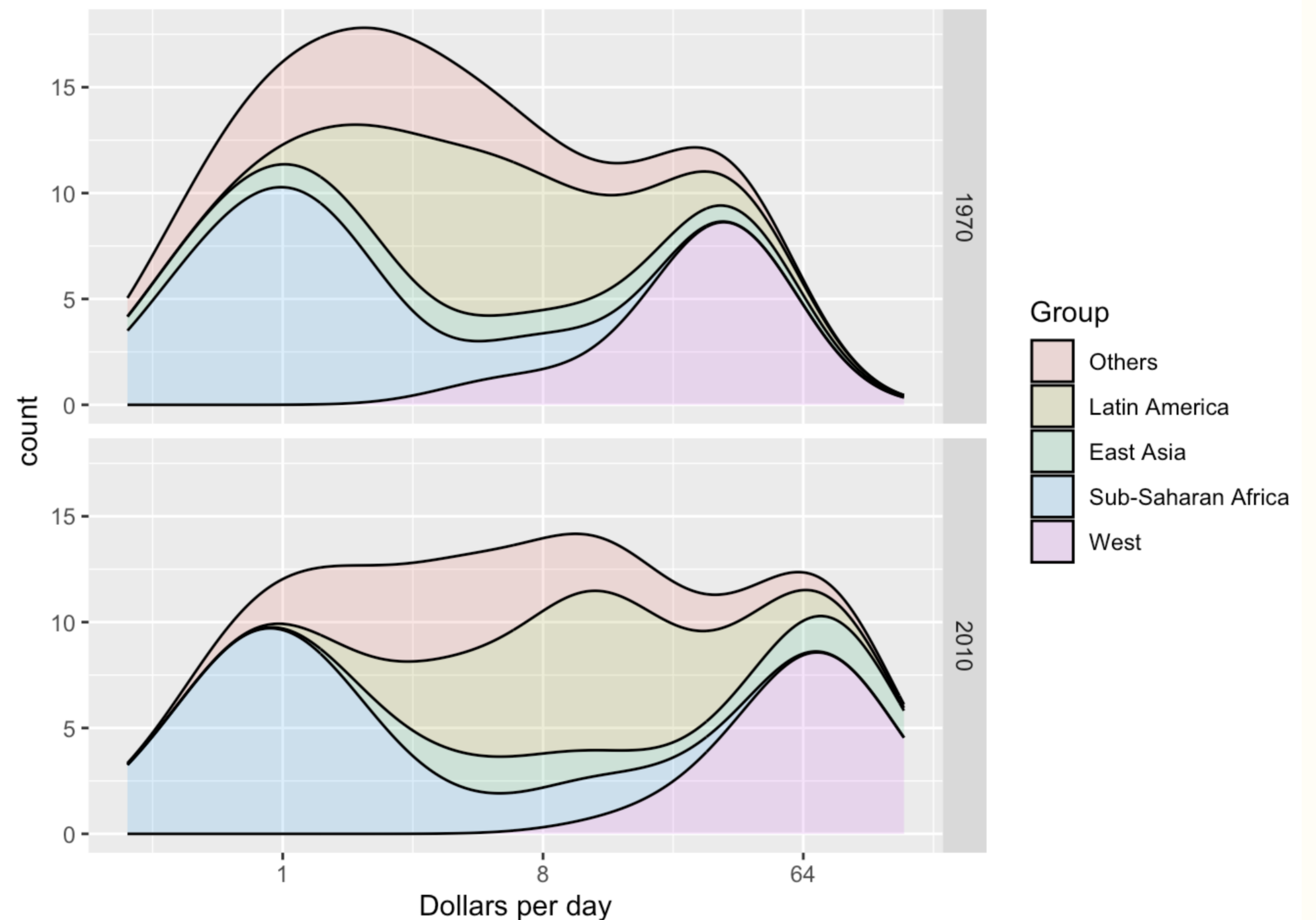
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



position = "stack"

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

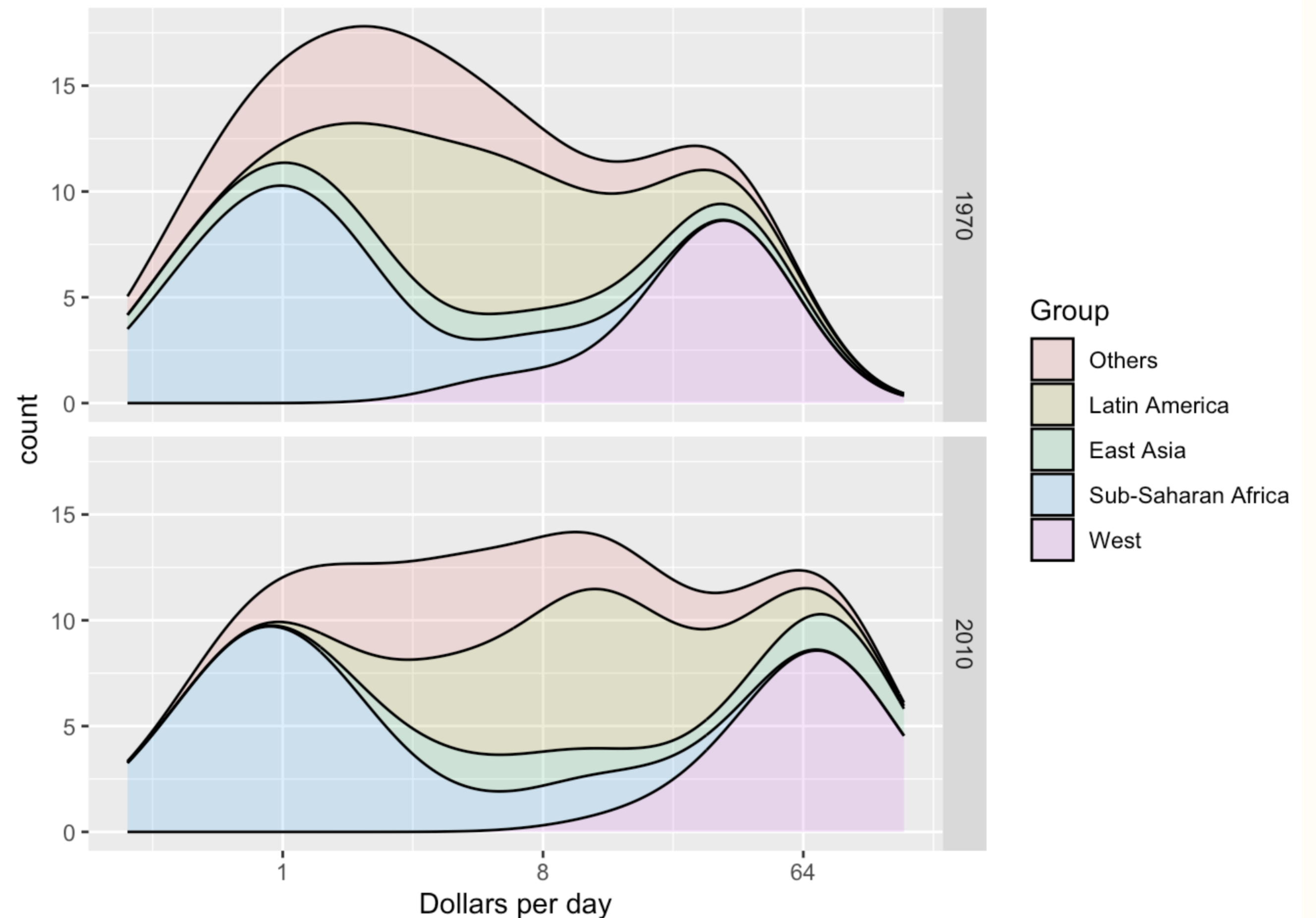
Plotting the two extremes (Sub-Saharan Africa and West) first shows the bimodality better



position = "stack"

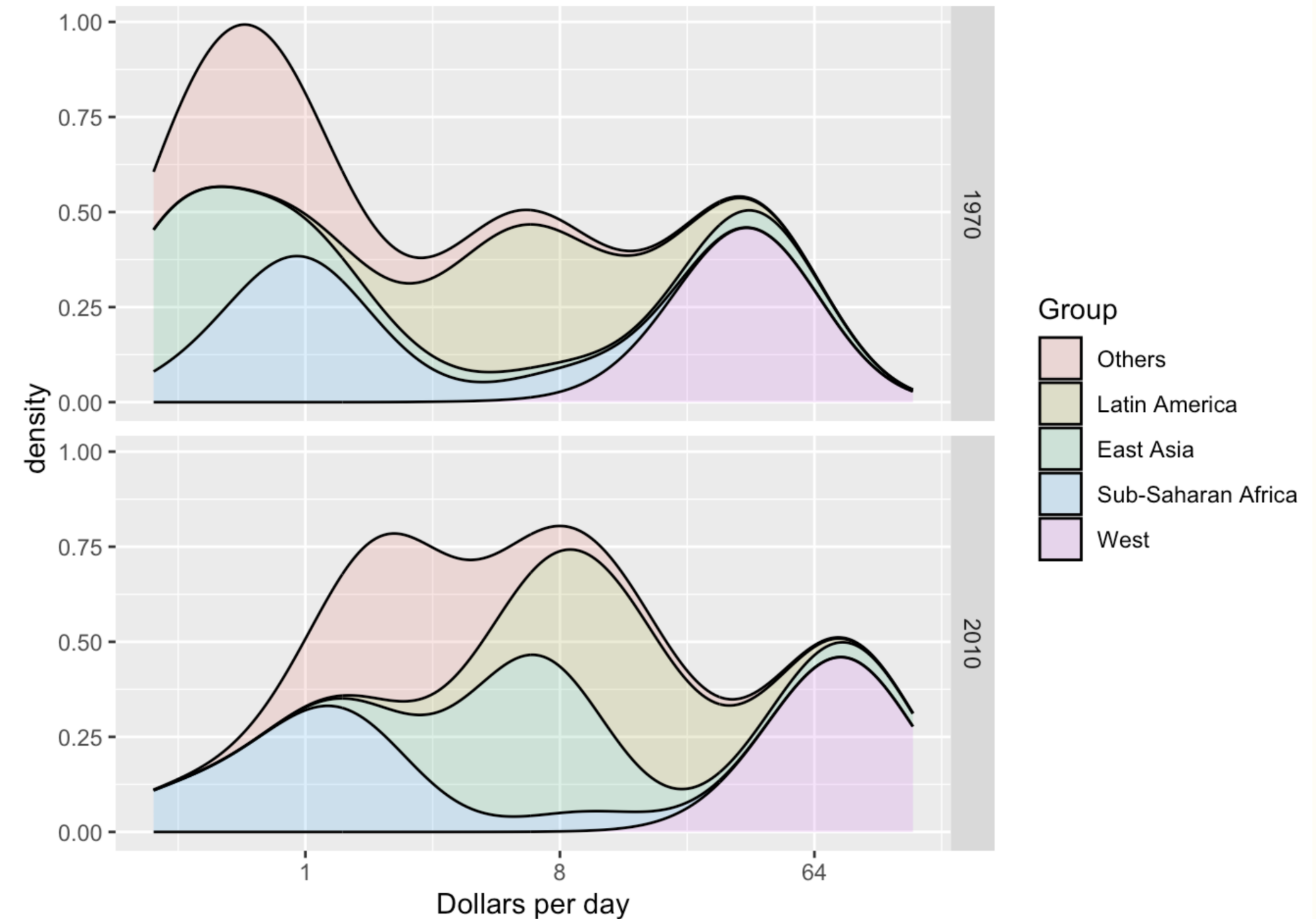
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

Here, each country has the same weight, which can skew comparisons – it's better to weight countries according to population



Weighted densities (**weight**)

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  group_by(year) %>%  
  mutate(weight = population/sum(population)) %>%  
  ungroup() %>%  
  ggplot(aes(dollars_per_day, fill = group, weight = weight)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



Summary

- Distributions across groups can be compared using smooth density plots
- The **case_when** function is an efficient way to create a categorical (factor) variable
- Smooth density plots can be stacked using the **position = "stack"** argument
- Smooth density plots can be weighted using the **weight** argument